

# Analyzing Student Performance with Machine Learning

## A Data Science Approach

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## 1 Introduction

This academic project explores how machine learning techniques can be applied to predict and understand student performance. Using student profile data including behavioral, academic, and contextual features, I aimed to identify patterns and assess predictive variables related to exam outcomes.

## 2 Dataset Overview

The primary dataset used includes 6,607 student records with 20 features. These features span:

- Academic Behavior: `Hours_Studied`, `Attendance`, `Previous_Scores`
- Environment and Context: `Motivation_Level`, `Parental_Involvement`, `Family_Income`
- Target Variable: `Exam_Score`

The data was sourced from multiple public student datasets on Kaggle. After evaluating the schemas, similar datasets were combined and standardized for consistency. This included aligning column names, converting categorical fields, and unifying formats. Missing values were handled using forward-fill and backward-fill strategies or by replacing with appropriate defaults. Categorical variables were encoded numerically to ensure compatibility with machine learning models. The cleaning process was fully automated through Python scripts for reproducibility.

## 3 Problem Statement

The goal of this project is to understand which factors most significantly influence student performance and to build predictive models for student outcomes (e.g., exam scores and GPA). I focused on analyzing behavioral, academic, and contextual indicators to build models that can generalize to unseen student records and help support data-informed academic interventions.

## 4 Exploratory Data Analysis

Initial analysis revealed key relationships:

- **Positive correlations:** `Attendance`, `Hours_Studied`, and `Previous_Scores` showed the strongest associations with `Exam_Score`.
- **Statistical Test:** A t-test comparing students with attendance above and below 80% yielded a statistically significant difference in exam performance ( $p < 0.001$ ).

I began my analysis by computing feature importances using a Random Forest Regressor implemented via `scikit-learn` to understand which features were most predictive of student performance. This revealed that `Attendance` and `Hours_Studied` were among the most important features. I supported this analysis by generating a correlation matrix using `pandas.corr()` and visu-

alized it with `seaborn.heatmap()` (Figure 3), as well as creating a feature importance bar graph (Figure 4).

To further examine key relationships, I used scatterplots generated with `seaborn.scatterplot()` and added best-fit regression lines using `numpy.polyfit()` and `matplotlib.pyplot.plot()` to visualize the relationship between `Attendance` and `Exam Score` (Figure 1), and `Hours Studied` and `Exam Score` (Figure 2). The respective slopes were 0.198 and 0.2893, indicating upward trends.

To statistically validate the impact of attendance on performance, I ran an independent two-sample t-test using `scipy.stats.ttest_ind()` to compare exam scores of students with less than 80% attendance to those with 80% or higher. The result was a p-value of less than 0.05, confirming that the difference in performance between low- and high-attendance groups was statistically significant.

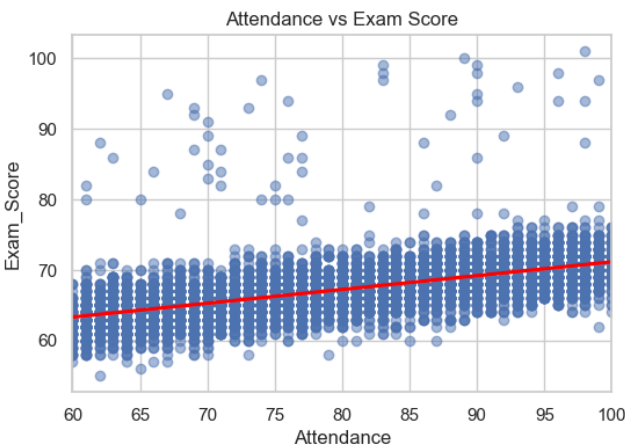


Figure 1: Attendance vs Exam Score

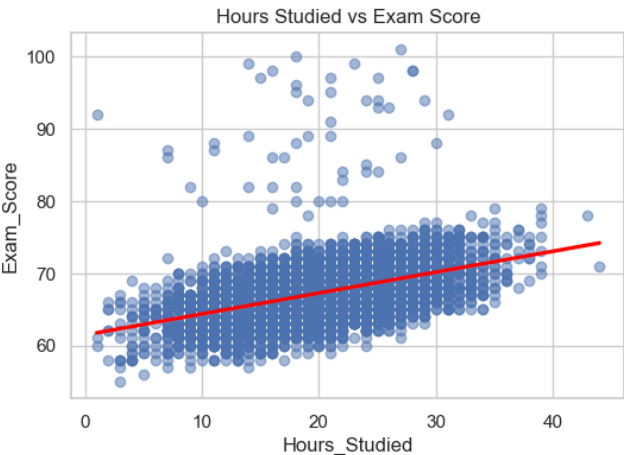


Figure 2: Hours Studied vs Exam Score

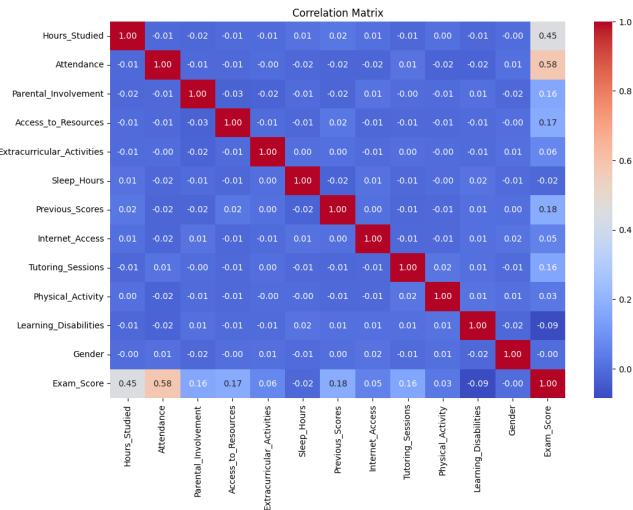


Figure 3: Correlation Matrix Heatmap

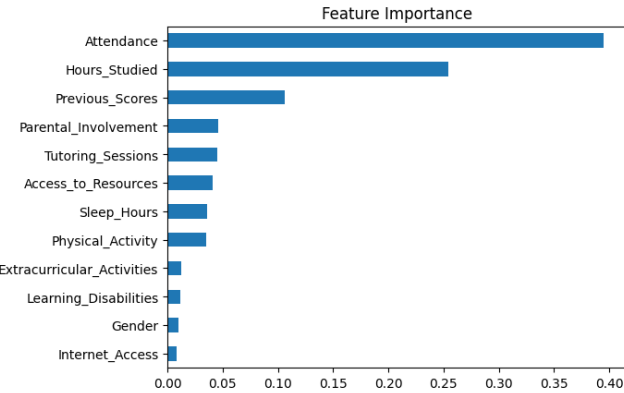


Figure 4: Feature Importances

## 5 Modeling and Prediction

Two supervised regression models were developed using Python’s `scikit-learn` library:

### 5.1 Linear Regression

- $R^2$ : 0.6680
- MAE: 1.16
- RMSE: 2.17

### 5.2 Gradient Boosting Regressor

- $R^2$ : 0.6448
- MAE: 1.27
- RMSE: 2.24
- Cross-validated  $R^2$ :  $0.5942 \pm 0.0551$

Linear Regression slightly outperformed the gradient model, indicating mostly linear relationships in the data.

### 5.3 Predicting GPA from Custom Input

To make the system practically usable, I built a regression model using three interpretable features: `StudyTimeWeekly`, `ParentalSupport`, and `Absences`. A Random Forest Regressor was trained using these features to predict GPA. I then created a minimal test file with the following input:

| Student | StudyTimeWeekly | ParentalSupport | Absences |
|---------|-----------------|-----------------|----------|
| 1       | 10.5            | 2               | 4        |
| 2       | 5.0             | 1               | 10       |
| 3       | 18.0            | 3               | 1        |

Table 1: Sample input used for GPA prediction using a Random Forest model

The model was able to predict GPAs for each student in the input file. The predictions were written to a new file, `predicted_gpa.csv`, which allows for downstream use in reporting, alerts, or integration with student dashboards.

To further assess model quality, I created a scatterplot comparing actual vs. predicted GPAs on the test set (Figure 5). This helped visualize how closely the predictions tracked with real data.

The final evaluation metrics for the Random Forest Regressor were as follows:

- $R^2$ : 0.8974

- MAE: 0.17
- MSE: 0.08
- RMSE: 0.28

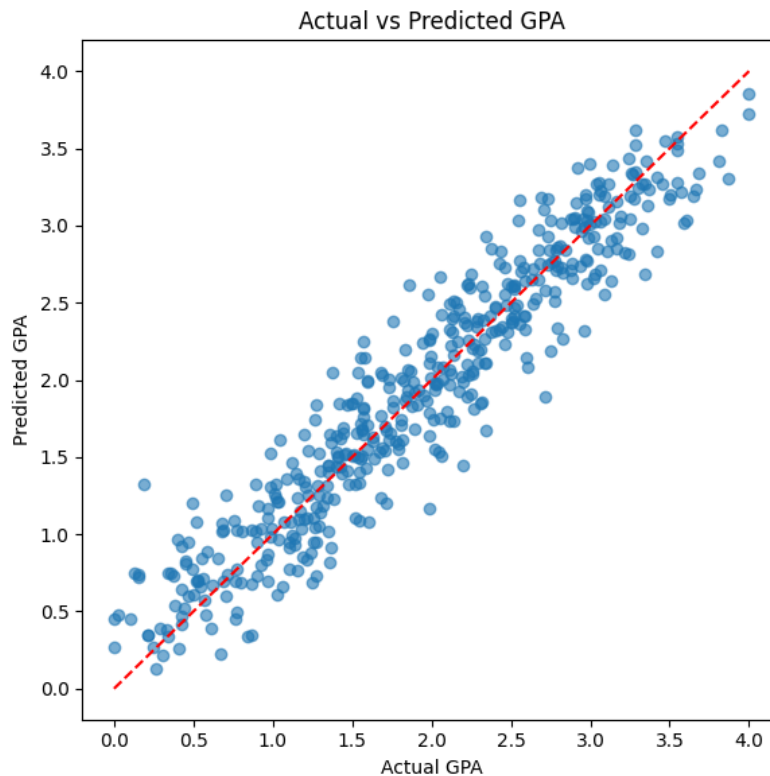


Figure 5: Actual vs Predicted GPA

This demonstrates how the trained model can be deployed on new student data to provide real-time predictions based on minimal inputs, supporting early academic interventions.

## 6 Unsupervised Learning: Clustering

I applied KMeans ( $k=3$ ) clustering using `scikit-learn`'s `KMeans` algorithm to identify behavioral profiles among students based on 8 features. Dimensionality reduction was applied using PCA to help visualize the clustering.

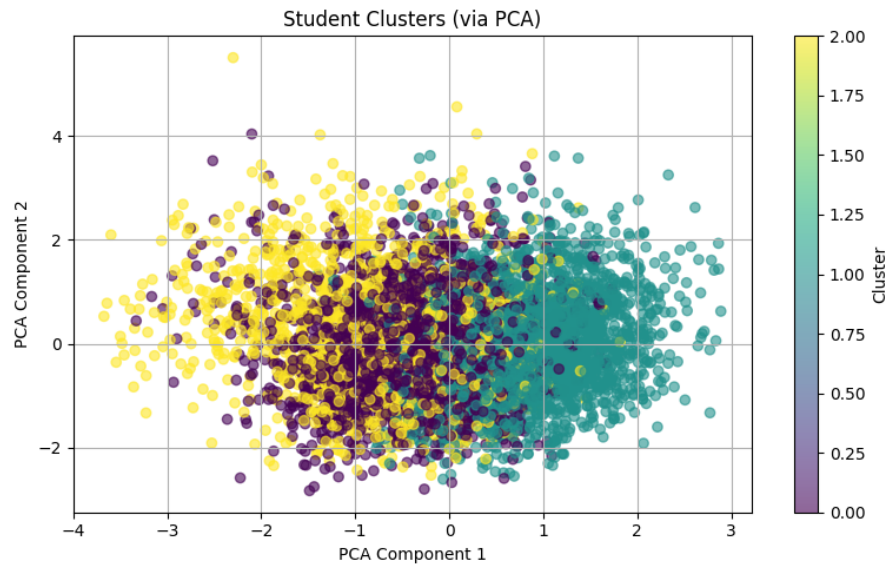


Figure 6: 2D PCA Visualization of Student Clusters

### Cluster Interpretations

- **Cluster 0:** Moderate attendance, high study time, some learning challenges
- **Cluster 1:** Very high attendance, reliable study habits, low learning disabilities — “Model Students”
- **Cluster 2:** Lower attendance, highest proportion of students with learning disabilities — “Support Needed”

## 7 Conclusion

This project demonstrated that attendance, hours studied, and historical performance are key drivers of student success. From a modeling perspective, linear regression yielded strong performance, slightly outperforming gradient boosting in terms of RMSE and interpretability. Random forest regressors provided flexible, robust predictions even with minimal features and proved practical for real-time GPA estimation.

Clustering allowed for unsupervised behavioral segmentation, uncovering three distinct groups of students with unique academic profiles. These insights could support personalized educational interventions, such as targeting tutoring or parental engagement based on cluster membership.

Furthermore, the statistical tests confirmed that attendance has a significant impact on student outcomes, with performance differences between low- and high-attendance groups being statistically significant.

The work also illustrates how predictive models can be operationalized by accepting structured CSV inputs and outputting tailored predictions. With modest extensions such as model interpretability

(e.g., SHAP values) and richer longitudinal data, this framework could serve as the foundation for a full-scale academic early warning system.

## 8 Limitations and Future Directions

While the project achieved promising results, a few limitations were encountered:

- **Limited Dataset Diversity:** The dataset, though rich in features, may not fully represent the variety found in different educational systems or student populations.
- **Model Interpretability:** Although Random Forests performed well, the explainability of the model's decisions is limited. Incorporating SHAP or LIME could address this.
- **Temporal Data:** The dataset lacked longitudinal tracking, limiting analysis of student progress over time.
- **If More Time Was Available:** I would implement SHAP value analysis for model interpretation, explore ensemble stacking for improved performance, and build a simple web app for real-time GPA prediction.
- **In Retrospect:** I would prioritize standardizing data earlier in the pipeline and automating more parts of the model evaluation process.

## 9 Project Experience Summary

**Project Title:** Student Performance Prediction with Machine Learning

**Tools and Technologies:** Python, pandas, scikit-learn, seaborn, matplotlib, LaTeX

**Overview:** Built a machine learning pipeline to analyze and predict student performance from a dataset of over 6,600 students. Developed models to forecast exam scores and GPA based on academic, behavioral, and contextual features.

### Key Contributions:

- Designed and implemented data cleaning scripts to preprocess and encode raw datasets from multiple sources.
- Built and evaluated regression models (Linear, Gradient Boosting, Random Forest) to predict exam scores and GPA.
- Created visualizations for feature correlations, importance, and model predictions.
- Conducted clustering analysis using PCA and KMeans to identify student behavior profiles.
- Automated GPA prediction for new student data using a standalone script and CSV input format.
- Authored full academic report and implemented statistical tests for performance validation.

**Outcome:** Achieved GPA prediction accuracy with  $R^2$  of 0.89 and exam score prediction accuracy of 0.66. Developed an end-to-end reproducible pipeline ready for academic early warning systems.

## References

- [1] Mahmoud Elhemaly. *Students Grading Dataset*. Kaggle, 2023. <https://www.kaggle.com/datasets/mahmoudehemaly/students-grading-dataset>
- [2] Lai Nguyen. *Student Performance Factors*. Kaggle, 2023. <https://www.kaggle.com/datasets/lainguyn123/student-performance-factors>
- [3] Rabie Elkhroua. *Students Performance Dataset*. Kaggle, 2023. <https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset>



Steven Bergner

To: Arshdeep Mann

Cc: CMPT 353 D1 <cmpt-353-help@sfu.ca>



Fri 2025-04-04 5:47 PM

Hi Arshdeep,

Thanks for sharing your project idea and for explaining your situation. I'm sorry to hear about your health challenges, and I appreciate the initiative you've shown in moving forward independently. From a course perspective, solo projects are allowed under special circumstances—just make sure to document your effort to find a team as outlined in the project instructions (e.g., forum post or email screenshot) and include that in your final submission.

Your proposed topic—predicting student performance based on sleep patterns—is both timely and well aligned with course goals. It's a great opportunity to apply classification and regression techniques in the context of behavioral and academic data. There are a couple of public datasets that support this kind of analysis: the [Student Performance dataset](#), which focuses on exam scores and demographic/behavioral features, and the [Student Sleep Patterns dataset](#), which includes information like sleep duration, bed and wake-up times, and GPA. These could serve as strong foundations for your project, or offer inspiration if you're collecting your own data.

Since you're working solo, I'd recommend starting with a core set of features—like sleep duration, bedtime consistency, and study hours—and a simple prediction target such as exam score or GPA. From there, you can expand if time allows. Even a focused, well-executed analysis will make for a strong final submission, especially if you include thoughtful visualization and interpretation of the results.

You're welcome to move forward with this project. Let me know if you'd like help narrowing your feature set or choosing the best modeling approach. Wishing you continued progress and a smooth recovery.

Best regards,

Steven

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Permission form signed by professor for independent project work.